

# A Comparative Study of Chinese-to-English Machine Translation Quality Based on Multi-dimensional Human Evaluation: DeepL, ChatGPT, Gemini, and Claude

## Abstract

This study looks at four common machine translation systems: DeepL, ChatGPT (GPT-4o), Google Gemini, and Claude. It tests their Chinese-to-English translation quality with 50 Chinese source texts from five text types: colloquial, news, academic, literary, and culture-loaded texts. In total, 200 translation outputs were collected. A five-dimensional human evaluation framework was used to score the translations from five aspects: accuracy, fluency, style matching, terminology accuracy, and cultural adaptation. The results show that the three large language model systems all performed better than the traditional neural machine translation system DeepL. Among them, Claude got the highest average score (4.79), followed by GPT and Gemini, while DeepL was lower (4.18). Text type had a clear influence on translation quality. The gap between systems was smaller in academic texts, but much larger in culture-loaded texts. Looking at the evaluation dimensions, the difference in fluency was already quite small. The main differences appeared in cultural adaptation and terminology accuracy. Also, each system had its own strengths in different text types, and no single system stayed ahead in all categories. The findings can give some reference for tool choice in translation practice, and they also add some empirical material to machine translation quality assessment studies in the age of large language models.

**Keywords:** machine translation; translation quality assessment; large language models; Chinese-to-English; DeepL; ChatGPT; Gemini; Claude

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## Chapter One Introduction

### 1.1 Research Background

Machine translation technology has developed quickly in the past ten years. In 2014, neural machine translation (NMT) models based on the encoder-decoder structure

appeared, and machine translation started to move from statistical methods to deep learning methods. After the Transformer architecture was proposed in 2017, translation quality improved again in an obvious way. DeepL is one of the more representative specialized NMT systems from this stage, and it has been widely used in translation practice.

After ChatGPT was released at the end of 2022, large language models (LLMs) quickly gained a lot of attention. Translation is not the only main task of these models, but their performance in translation caused a lot of discussion. Later, Google released Gemini and Anthropic released Claude. These general large models also became important in translation tasks. Compared with traditional NMT, LLMs show some advantages in world knowledge, context understanding, and flexible language generation. This has also changed the competition among machine translation tools.

In China, the demand for Chinese-to-English translation has always been high. For example, academic paper publication, international news communication, literary translation, and cross-cultural communication all need good English translations. Compared with this demand, systematic evaluation studies on the new generation of LLM systems in Chinese-to-English translation are still not very many. Studies that include multiple text types and multiple evaluation dimensions at the same time are especially limited. So, it is necessary to make a more specific comparison of the current mainstream systems.

## **1.2 Research Questions**

This study mainly focuses on the following three questions:

RQ1: How can the overall translation quality of DeepL, ChatGPT, Gemini, and Claude be ranked in Chinese-to-English translation?

RQ2: Under five text types, namely colloquial, news, academic, literary, and culture-loaded texts, do these systems show obvious differences? Which text type can best separate the systems in quality?

RQ3: In the five dimensions of accuracy, fluency, style matching, terminology accuracy, and cultural adaptation, what are the strengths and weaknesses of each system?

## **1.3 Significance of the Study**

The significance of this study can be seen in three main aspects.

First, at the theoretical level, research on Chinese-to-English machine translation quality in the age of large language models is still not enough, especially studies based on

multi-dimensional human evaluation. This paper hopes to add some empirical material in this area.

Second, at the practical level, different translation tasks do not ask for exactly the same kind of translation. By comparing different text types and different systems, this study can offer some reference for translators, language service companies, and foreign language teaching when they choose tools.

Third, at the methodological level, this paper builds a five-dimensional evaluation framework that includes cultural adaptation. It tries to be more complete than evaluation methods that only depend on accuracy and fluency. It may also offer some help for similar future studies.

## **1.4 Structure of the Thesis**

This thesis has six chapters. Chapter One is the introduction, which presents the research background, research questions, and significance of the study. Chapter Two is the literature review, which looks back at the development of machine translation, evaluation methods, and related research. Chapter Three is the research methodology, which explains the data sources, system settings, and evaluation framework. Chapter Four presents the research results, including major data and cases. Chapter Five is the discussion, where the results are analyzed and explained. Chapter Six is the conclusion, which summarizes the main findings, points out the limitations, and gives suggestions for future research.

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# **Chapter Two Literature Review**

## **2.1 The Development of Machine Translation**

### **2.1.1 Rule-Based Machine Translation (1950s–1990s)**

Machine translation can be traced back to the Russian-English translation experiment carried out by Georgetown University and IBM in 1954. Early systems mainly depended on grammar rules and bilingual dictionaries written by humans. A typical example is SYSTRAN in 1968. This kind of rule-based machine translation (RBMT) worked to some degree in limited fields, but in open-domain texts, translation quality was often unstable because the rule coverage was limited.

### **2.1.2 Statistical Machine Translation (1990s–2014)**

After Brown et al. (1990) proposed statistical machine translation (SMT), machine translation entered a new stage. SMT used large-scale parallel corpora to learn translation probabilities, so it reduced the dependence on hand-written rules. In the 2000s, phrase-based statistical machine translation gradually became mainstream, and early Google Translate also used this technical route. Still, SMT had clear limits when dealing with long-distance dependency and language pairs with large word-order differences.

### **2.1.3 Neural Machine Translation (2014–present)**

Sutskever et al. (2014) proposed the sequence-to-sequence model based on the encoder-decoder structure, which marked the start of the neural machine translation era. Later, Bahdanau et al. (2015) introduced the attention mechanism, and this further improved translation quality. After Vaswani et al. proposed the Transformer architecture in 2017, neural machine translation entered a stage of rapid development. Most mainstream translation systems today are related to the Transformer architecture at the basic level. DeepL is often seen as one of the representative specialized NMT systems from this stage.

### **2.1.4 The Era of Large Language Models (2022–present)**

After ChatGPT was released in November 2022, the general language processing ability of large language models got wide attention. Translation is only one of their many abilities, but in real use it has shown strong competitiveness. From 2023 to 2024, GPT-4, GPT-4o, Google Gemini, and Anthropic Claude were released one after another. Because of this, machine translation research started to pay more attention to the differences between LLMs and traditional NMT. These models are usually trained on large-scale multilingual data, so they show new features in knowledge storage, reasoning ability, and context understanding.

## **2.2 Introduction to the Four Translation Systems**

**DeepL:** DeepL was released by DeepL GmbH in Germany. It is a representative specialized translation system. Its advantages are usually fast translation speed and stable output, and it has high popularity in real use.

**ChatGPT (GPT-4o):** ChatGPT is a general large language model developed by OpenAI. GPT-4o is one of its important versions. It has strong language understanding and generation ability, and translation is one of its application areas.

**Google Gemini:** Gemini was developed by Google DeepMind. Since Google has long-term experience in machine translation and multilingual processing, Gemini also shows strong potential in translation.

**Claude:** Claude was developed by Anthropic. It puts more focus on safety and text understanding, and it has received attention for long-text processing and careful language expression.

## 2.3 Machine Translation Quality Evaluation Methods

### 2.3.1 Automatic Evaluation Metrics

BLEU (Papineni et al., 2002) is one of the most common automatic evaluation metrics for machine translation. It mainly scores translation quality by calculating n-gram overlap between machine output and reference translation. Later, other metrics such as METEOR (Banerjee & Lavie, 2005), TER (Snover et al., 2006), and COMET (Rei et al., 2020) were also proposed.

Still, automatic metrics have some problems. First, they usually depend on reference translations, but translation often allows more than one reasonable version. Second, for deeper aspects of translation quality, such as cultural adaptation and register matching, automatic metrics are not very good at giving accurate judgments. Third, in language pairs like Chinese and English, where word order differs a lot, surface matching and human judgment are not always highly consistent.

### 2.3.2 Human Evaluation Methods

Human evaluation is still widely seen as an important method in machine translation quality assessment. Common methods mainly include the following:

- (1) **Adequacy and fluency evaluation:** This method checks whether the translation fully conveys the source meaning and whether the target language expression is natural and smooth.
- (2) **Direct Assessment (DA):** Evaluators give an overall score to the translation on a continuous scale (Graham et al., 2013).
- (3) **MQM (Multidimensional Quality Metrics):** This method labels and classifies translation problems based on error types (Lommel et al., 2014).

This study uses a five-dimensional human evaluation framework. Besides accuracy and fluency, it also adds style matching, terminology accuracy, and cultural adaptation, so that the differences between systems can be observed more fully.

## 2.4 Current Research on Chinese-to-English Machine Translation

In recent years, more studies have appeared on Chinese-to-English machine translation quality evaluation. However, many of them focus mainly on news texts and academic

texts, and they often rely on automatic metrics such as BLEU. Colloquial texts, literary texts, and online cultural expressions have received less attention.

As for LLM translation research, Jiao et al. (2023) made an early evaluation of ChatGPT's translation ability and pointed out that it was already close to commercial NMT systems in some high-resource language pairs. Hendy et al. (2023) later carried out a more systematic study on GPT-4's translation performance. However, studies that include Claude and Gemini at the same time and focus especially on Chinese-to-English translation across multiple text types are still limited.

## 2.5 Text Types and Translation Difficulty

According to Reiss (1971, 1976), different text types differ in function and expression, so they also ask for different translation strategies. Based on this idea, this study divides the test data into five types:

**Colloquial texts:** These are more spoken, often emotional, and often contain implied meaning and pragmatic information.

**News texts:** These stress full information, standard expression, and formal register.

**Academic texts:** These stress logic, accurate terminology, and careful expression.

**Literary texts:** These pay more attention to rhetoric, imagery, and style.

**Culture-loaded texts:** These include idioms, two-part allegorical sayings, internet slang, and cultural allusions, and they require strong ability in cultural understanding and transfer.

The domestication and foreignization strategies proposed by Venuti (1995) are also useful for translating culture-loaded content.

## 2.6 Research Gaps and the Position of This Study

In general, current research still has several limits. First, there are not enough systematic comparisons across multiple text types. Second, newer LLM systems such as Claude and Gemini have not been discussed enough. Third, in human evaluation, cultural adaptation is often not treated as an independent dimension. This paper hopes to add to these areas through multi-system, multi-text-type, and multi-dimensional evaluation.

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# Chapter Three Research Methodology

## 3.1 Research Design

This study uses a comparative experiment design. Four translation systems, five text types, and ten source texts in each type together produced 200 translation outputs. Then, these translations were scored based on five evaluation dimensions. The results were used to compare different systems in overall performance, performance across text types, and performance across evaluation dimensions.

### 3.2 Data Selection

This study selected 50 Chinese source texts in total. They were divided into five text types, with 10 texts in each type.

**Colloquial texts (Colloquial, C01–C10):** These were taken from daily conversations, film or TV lines, and social media expressions. They include features such as irony, urging, and emotional expression.

**News texts (News, N01–N10):** These were taken from mainstream news reports and cover science, sports, culture, and politics.

**Academic texts (Academic, A01–A10):** These were taken from Chinese academic abstracts in fields such as natural language processing, psycholinguistics, and translation studies.

**Literary texts (Literary, L01–L10):** These were selected from works by writers such as Lu Xun, Zhu Ziqing, Lao She, Yu Hua, and Zhang Ailing.

**Culture-loaded texts (Cultural, K01–K10):** These include internet slang, common sayings, two-part allegorical sayings, and idioms.

The data selection followed three main principles.

First, representativeness: each type of text should show common translation difficulties in that category.

Second, some difference in difficulty: this helps avoid data that is too simple or too single.

Third, comparability: the length of each item was controlled as much as possible within one or two sentences.

Table 1 lists the 50 Chinese source texts, so they are not repeated one by one in the main body here.

#### Table 1 Chinese Source Texts (50 Items)

| ID  | Type       | Chinese Source Text | Source                                           | Difficulty Tags                                  |
|-----|------------|---------------------|--------------------------------------------------|--------------------------------------------------|
| C01 | Colloquial | 道歉有用的话，要警察干嘛？       | <i>Meteor Garden</i> script page                 | rhetorical question; irony; spoken tone          |
| C02 | Colloquial | 想哭的时候倒立，眼泪就流不出来了。   | <i>Meteor Garden</i> script page                 | spoken tone; fixed expression                    |
| C03 | Colloquial | 我要我觉得，不要你觉得！        | famous variety show moment by Huang Xiaoming     | spoken tone; spoken rhythm; forceful tone        |
| C04 | Colloquial | 你有病啊，当然不是我的毛。       | Douban transcript of <i>The Mermaid</i>          | spoken tone; rhetorical question; blunt register |
| C05 | Colloquial | 你是不是傻？这么明显的事情你看不出来？ | common daily spoken expression                   | rhetorical question; spoken tone; blunt register |
| C06 | Colloquial | 滑雪就像人生一样不是吗……       | Douban line from <i>The King of Winter Sleep</i> | spoken tone; rhetorical question; metaphor       |
| C07 | Colloquial | 我真是谢谢你了，帮我帮成这样。     | daily ironic spoken expression                   | irony; spoken tone; pragmatic meaning            |
| C08 | Colloquial | 你看了几本上野千鹤子啊。        | Douban quotation from <i>Her Story</i>           | spoken tone; sarcasm; cultural proper name       |



| ID  | Type       | Chinese Source Text           | Source                          | Difficulty Tags                                 |
|-----|------------|-------------------------------|---------------------------------|-------------------------------------------------|
| C09 | Colloquial | 这一天天的，烦死了，我不演了，直接摊牌。          | internet spoken expression      | internet speech; emotional tone; spoken rhythm  |
| C10 | Colloquial | 你别搁这闲聊了，快走吧！                  | Zhao Benshan sketch dialogue    | spoken tone; spoken rhythm; dialect flavor      |
| N01 | News       | 生成式人工智能正在助力新闻进入“液态内容”阶段。      | The Paper                       | news register; metaphorical term                |
| N02 | News       | 体重管理并非越瘦越好，而是将体重维持在健康、适宜的范围内。 | People's Daily                  | news register; contrast structure               |
| N03 | News       | 制造业是立国之本、强国之基。                | Xinhua News Agency              | news register; formal style; political rhetoric |
| N04 | News       | 北京时间2月7日凌晨，米兰冬奥会即将拉开帷幕。       | People's Daily Overseas Edition | news register; time expression; proper noun     |
| N05 | News       | 中文热是贴近中                       | People's Daily                  | news register; metaphorical                     |

| ID  | Type | Chinese Source Text                             | Source             | Difficulty Tags                                       |
|-----|------|-------------------------------------------------|--------------------|-------------------------------------------------------|
|     |      | 国、读懂中国的必然产物，更是中国国际影响力与日俱增的生动注脚。                 |                    | wording;<br>formal style                              |
| N06 | News | 学中文、用中文，还只是世界友人了解中国、亲近中国浪潮里的一朵浪花。               | People's Daily     | news register;<br>literary metaphor;<br>formal style  |
| N07 | News | 当地时间7日，中国队组合王子菲/盛李豪在射击世界杯西班牙站10米气步枪混合团体决赛中逆转夺冠。 | People's Daily     | news register;<br>dense information;<br>proper noun   |
| N08 | News | 体育，承载着国家强盛、民族振兴的梦想。                             | People's Daily     | news register;<br>political rhetoric;<br>formal style |
| N09 | News | 我国智能机器人产                                        | Xinhua News Agency | news register;<br>industrial                          |

| ID  | Type     | Chinese Source Text                                  | Source             | Difficulty Tags                                           |
|-----|----------|------------------------------------------------------|--------------------|-----------------------------------------------------------|
|     |          | 业创新能力持续提升，应用场景不断拓展。                                  |                    | policy language; formal style                             |
| N10 | News     | 从“跟跑”“并跑”再到“领跑”，中国用几十年时间走完了西方发达国家几百年走过的工业化历程。        | Xinhua News Agency | news register; metaphorical wording; historical narrative |
| A01 | Academic | 文本倾向分析目的是确定文本所表达的态度或观点，近几年来已经成为信息检索和自然语言处理领域的一个热点问题。 | CNKI abstract      | technical term; academic abstract style                   |
| A02 | Academic | 阅读知觉广度通常指读者在阅读文本过程中每次注视能获取有用视觉信                      | CNKI abstract      | technical term; definition structure                      |

| ID  | Type     | Chinese Source Text                                              | Source        | Difficulty Tags                            |
|-----|----------|------------------------------------------------------------------|---------------|--------------------------------------------|
|     |          | 息的范围。                                                            |               |                                            |
| A03 | Academic | 采用眼动追踪技术探讨中文阅读中无关言语效应的影响机制。                                      | CNKI abstract | technical term;<br>academic abstract style |
| A04 | Academic | 虽然神经机器翻译模型使用大规模数据集进行训练能够改善翻译模型的表现，但是数据集中有关句子内容类别以及结构的信息并未得到充分利用。 | CNKI abstract | technical term;<br>long complex sentence   |
| A05 | Academic | 议论文自动生成是自然语言生成中一项极具挑战性的任务。                                       | CNKI abstract | technical term;<br>academic abstract style |
| A06 | Academic | 自动词语简化是用简单、同等意义的                                                 | CNKI abstract | technical term;<br>definition structure    |

| ID  | Type     | Chinese Source Text                     | Source        | Difficulty Tags                            |
|-----|----------|-----------------------------------------|---------------|--------------------------------------------|
|     |          | 词语替代句子中复杂词的过程，是文本简化中的一个重要研究方向。          |               |                                            |
| A07 | Academic | 从 2017 年以来，自然语言处理中提出了“预训练+微调+师生学习”的新范式。 | CNKI abstract | technical term;<br>academic abstract style |
| A08 | Academic | 机器翻译凭借着速度和效率上的优势，在语言服务行业发挥着越来越重要的作用。    | CNKI abstract | technical term;<br>academic abstract style |
| A09 | Academic | 自从机器翻译诞生以来，机器翻译能否取代人工翻译这一疑问一直存在。        | CNKI abstract | technical term;<br>academic abstract style |

| ID  | Type     | Chinese Source Text                                        | Source                           | Difficulty Tags                                 |
|-----|----------|------------------------------------------------------------|----------------------------------|-------------------------------------------------|
| A10 | Academic | 由于机器翻译的机械性和随机性等短板，机器翻译无法完全取代人工翻译，机器翻译和人工翻译将长期共存并相互补充、相互促进。 | CNKI abstract                    | technical term;<br>long complex sentence        |
| L01 | Literary | 我翻开历史一查，这历史每叶上都写着“仁义道德”几个字。                                | Lu Xun, <i>Diary of a Madman</i> | literary rhetoric; early vernacular Chinese     |
| L02 | Literary | 仔细看了半夜，才从字缝里看出字来，满本都写着两个字，是“吃人”。                           | Lu Xun, <i>Diary of a Madman</i> | literary rhetoric;<br>metaphor;<br>shock effect |
| L03 | Literary | 我与父亲不相见已有二年余了，我最不能忘记的是他的背影。                                | Zhu Ziqing, <i>Back View</i>     | literary rhetoric;<br>emotional expression      |

| ID  | Type     | Chinese Source Text         | Source                                           | Difficulty Tags                                                |
|-----|----------|-----------------------------|--------------------------------------------------|----------------------------------------------------------------|
| L04 | Literary | 人是为活着本身而活着，不是为活着之外的任何事物所活着。 | Yu Hua, <i>To Live</i>                           | literary<br>rhetoric;<br>philosophical<br>expression           |
| L05 | Literary | 这几天心里颇不宁静。                  | Zhu Ziqing, <i>Moonlight over the Lotus Pond</i> | literary<br>rhetoric;<br>emotional<br>expression               |
| L06 | Literary | 生命是一袭华美的袍，爬满了蚤子。            | Zhang Ailing, <i>Genius Dream</i>                | literary<br>rhetoric;<br>metaphor;<br>aesthetic<br>effect      |
| L07 | Literary | 他没有什么模样，使他可爱的是脸上的精神。        | Lao She, <i>Rickshaw Boy</i>                     | literary<br>rhetoric;<br>character<br>description              |
| L08 | Literary | 他确乎有点象一棵树，坚壮，沉默，而又有生气。      | Lao She, <i>Rickshaw Boy</i>                     | literary<br>rhetoric;<br>metaphor;<br>character<br>description |
| L09 | Literary | 过去的日子如轻烟，被微风吹散了，如薄雾，被初阳蒸融了。 | Zhu Ziqing, <i>Hurry</i>                         | literary<br>rhetoric;<br>parallelism;<br>imagery               |

| ID  | Type     | Chinese Source Text                             | Source                                            | Difficulty Tags                                          |
|-----|----------|-------------------------------------------------|---------------------------------------------------|----------------------------------------------------------|
| L10 | Literary | 盼望着，<br>盼望着，<br>东风来了，春天的脚步近了。                   | Zhu<br>Ziqing, <i>Spring</i>                      | literary<br>rhetoric;<br>repetition;<br>personification  |
| K01 | Cultural | 不作死就不会死。                                        | People.cn<br>quotation of<br>online<br>expression | internet slang;<br>culture-loaded                        |
| K02 | Cultural | 一清二白。                                           | People's Daily<br>Overseas Edition                | idiom; culture-loaded                                    |
| K03 | Cultural | 外甥打灯笼——照旧（舅）。                                   | China Culture<br>Research<br>Institute            | two-part<br>allegorical<br>saying;<br>homophonic<br>pun  |
| K04 | Cultural | 我吃过的盐比你吃过的米还多。                                  | People's Daily<br>Overseas Edition                | saying;<br>culture-loaded                                |
| K05 | Cultural | 世界那么大，我想去看看。                                    | China.org.cn                                      | internet<br>catchphrase;<br>culture-loaded               |
| K06 | Cultural | 比如，大量数字、符号、字母混合使用，“3Q”表示谢谢你，“9494”表示就是就是，“emo”表 | Xinhua News<br>Agency                             | internet<br>language;<br>abbreviation;<br>culture-loaded |



| ID  | Type     | Chinese Source Text | Source             | Difficulty Tags                        |
|-----|----------|---------------------|--------------------|----------------------------------------|
|     |          | 示难过，“YYDS”意思是永远的神。  |                    |                                        |
| K07 | Cultural | 破防。                 | Xinhua News Agency | internet slang; culture-loaded         |
| K08 | Cultural | 我看不懂，但我大受震撼。        | Xinhua News Agency | internet slang; culture-loaded         |
| K09 | Cultural | 伤害性不高，侮辱性极强。        | Xinhua News Agency | internet slang; rhetorical parallelism |
| K10 | Cultural | 尸位素餐。               | People.cn          | idiom; culture-loaded                  |

### 3.3 Translation System Settings

The outputs of the four translation systems were all collected on April 18, 2025, in order to reduce the effect of version differences as much as possible. During translation, the default settings were used. No extra complex prompts were added. The Chinese source text was entered directly and translated into English.

**Table 2 Translation System Settings**

| System        | Version/Model | Collection Method          |
|---------------|---------------|----------------------------|
| DeepL         | Web version   | paste source text directly |
| ChatGPT       | GPT-4o        | input in chat interface    |
| Google Gemini | Gemini        | input in chat interface    |
| Claude        | Claude        | input in chat interface    |

### 3.4 Evaluation Framework

This paper builds a five-dimensional human evaluation framework. Each dimension was scored with a 1–5 Likert scale.

**Table 3 Five-dimensional Evaluation Framework**

| Dimension           | Definition                                                                 | Scoring Standard                                 |
|---------------------|----------------------------------------------------------------------------|--------------------------------------------------|
| Accuracy            | how faithfully the translation conveys the source meaning                  | 5 = completely accurate, 1 = seriously distorted |
| Fluency             | how natural and smooth the translation is in the target language           | 5 = fully natural, 1 = hard to understand        |
| Style               | how well the translation matches the register and style of the source text | 5 = perfect match, 1 = serious mismatch          |
| Terminology         | how correctly technical terms and fixed expressions are translated         | 5 = completely correct, 1 = serious error        |
| Cultural Adaptation | how well culture-loaded content is handled                                 | 5 = perfect adaptation, 1 = complete failure     |

The overall score (Avg\_Score) is the arithmetic mean of the five dimension scores.

### 3.5 Evaluation Procedure

The scoring was completed by the researcher. During the evaluation process, all translations were checked one by one against the source text, and the main reasons for scoring were recorded. After all the scoring was finished, the data were checked again to make sure the scores were generally consistent and the statistics and average scores were correct.

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## Chapter Four Research Results

### 4.1 Overall Performance of the Systems

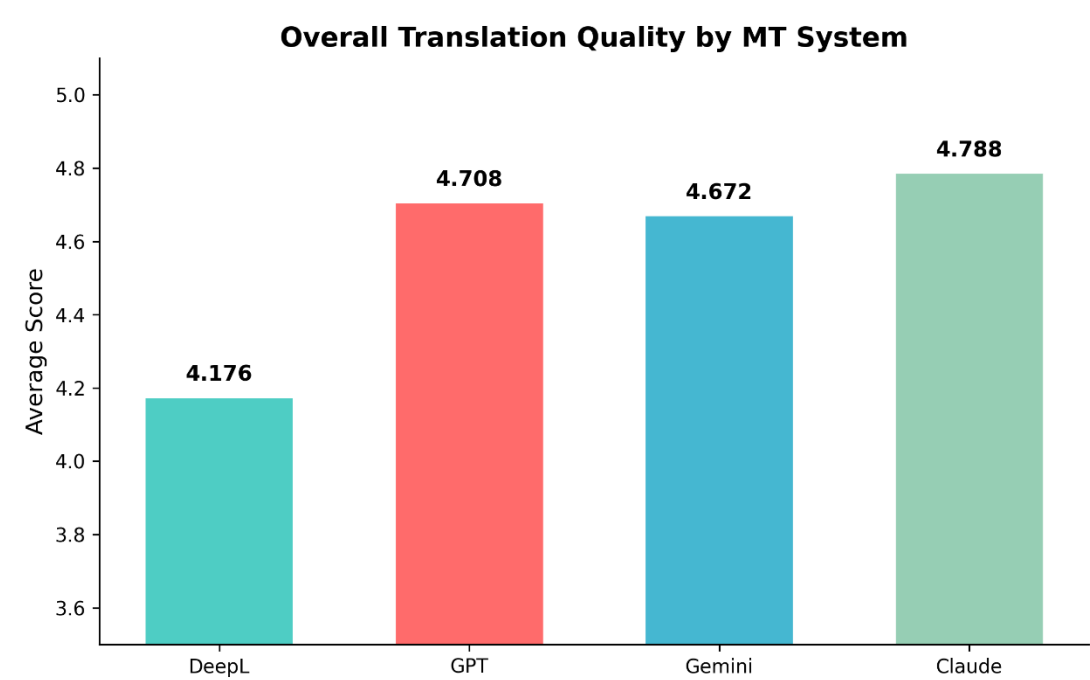


Figure 1. Overall average translation quality scores of four MT systems.

**Table 4 Ranking of Overall Average Scores of the Four Systems**

| Rank | System | Average Score |
|------|--------|---------------|
| 1    | Claude | 4.788         |
| 2    | GPT    | 4.708         |
| 3    | Gemini | 4.672         |
| 4    | DeepL  | 4.176         |

From the overall average scores, the three LLM systems all performed clearly better than DeepL. Claude ranked first with 4.788. GPT and Gemini ranked second and third, and both of them scored above 4.67. By comparison, DeepL got 4.176, which shows a visible gap from the three LLMs. Since this study uses a 1–5 scale, this gap is already meaningful. It suggests that there is a fairly clear quality difference between these types of systems.

### 4.2 Performance Across Different Text Types

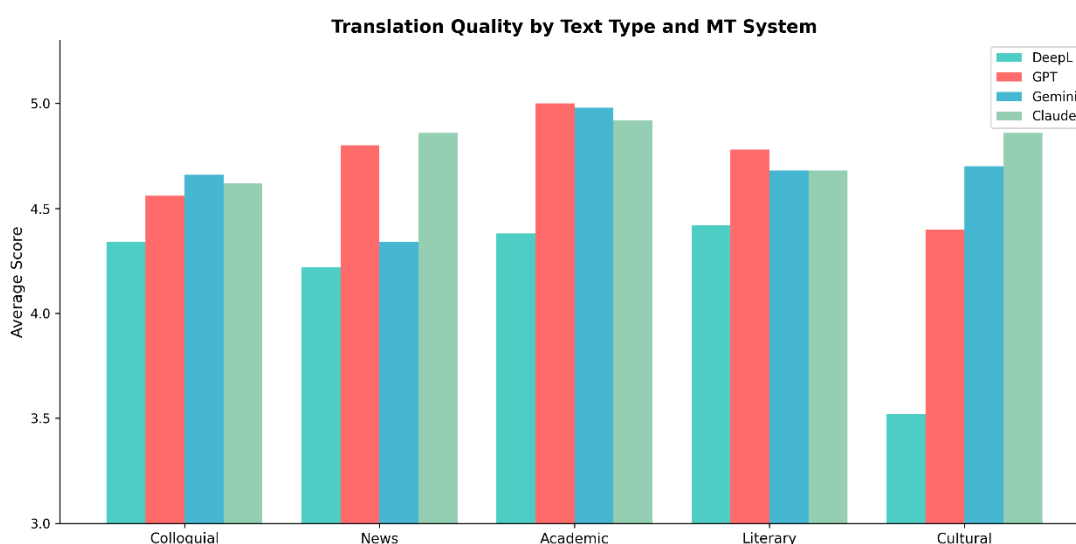


Figure 2. Average scores by text type and MT system.

**Table 5 Average Scores of the Four Systems Across Text Types**

| Text Type  | DeepL | GPT  | Gemini | Claude | Best in This Type |
|------------|-------|------|--------|--------|-------------------|
| Colloquial | 4.34  | 4.56 | 4.66   | 4.62   | Gemini            |
| News       | 4.22  | 4.80 | 4.34   | 4.86   | Claude            |
| Academic   | 4.38  | 5.00 | 4.98   | 4.92   | GPT               |
| Literary   | 4.42  | 4.78 | 4.68   | 4.68   | GPT               |
| Cultural   | 3.52  | 4.40 | 4.70   | 4.86   | Claude            |

Looking at text types, academic texts got the highest scores overall. All four systems scored highly in this type, and GPT even reached a full score of 5.00. This suggests that academic texts, with more regular terminology and sentence patterns, are relatively friendly for machine translation systems.

Culture-loaded texts showed the largest differences between systems. DeepL only got 3.52 in this type, while Claude reached 4.86. The gap between them was 1.34 points, which was the biggest among all text types. This suggests that cultural knowledge, pragmatic understanding, and expression transfer have a strong effect on translation quality, and these are exactly the places where different systems are easiest to separate.

Also, the systems did not have the same strong areas. GPT did better in academic and literary texts. Gemini got the highest score in colloquial texts. Claude did best in news and culture-loaded texts. This shows that in real use, we should not simply think one system fits every task. The choice should depend on the text type.

The differences among the three LLMs in colloquial texts were small, only from 4.56 to 4.66. This means that in common daily expressions and spoken tone, the major large models are already quite close. By contrast, Gemini performed worse than the other two LLMs in news texts. This suggests that it still has room to improve in formal news register and some political rhetorical expressions.

### 4.3 Performance Across Evaluation Dimensions

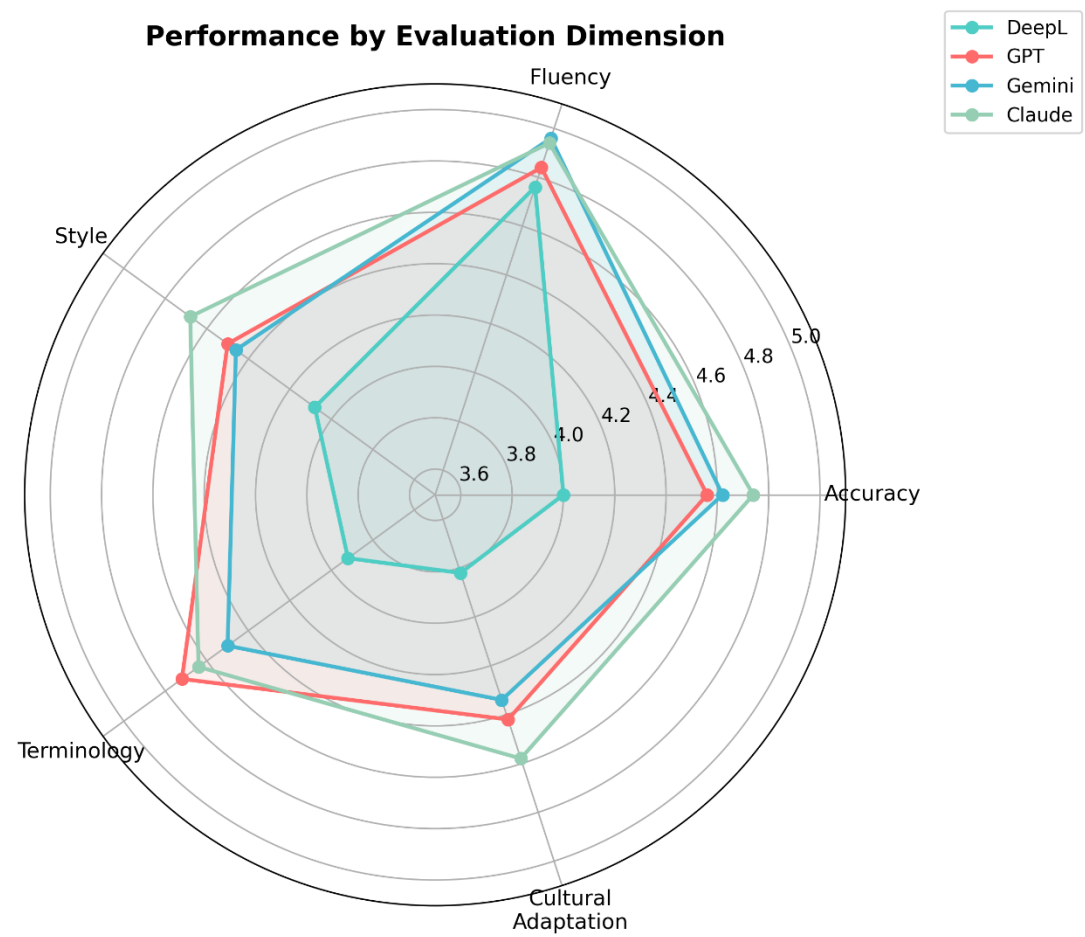


Figure 3. Radar chart of performance across five evaluation dimensions.

**Table 6 Average Scores of the Four Systems Across Evaluation**

**Dimensions**

| Dimension           | DeepL | GPT  | Gemini | Claude |
|---------------------|-------|------|--------|--------|
| Accuracy            | 4.00  | 4.56 | 4.62   | 4.74   |
| Fluency             | 4.76  | 4.84 | 4.96   | 4.94   |
| Style               | 4.08  | 4.50 | 4.46   | 4.68   |
| Terminology         | 3.92  | 4.72 | 4.50   | 4.64   |
| Cultural Adaptation | 3.82  | 4.42 | 4.34   | 4.58   |

From the results of different dimensions, fluency is no longer the main factor that separates the systems. All four systems got high scores in this dimension, and the difference was only 0.20 points. This shows that producing natural and readable English has already become a basic ability of current machine translation systems.

By contrast, cultural adaptation and terminology accuracy show clearer differences between systems. In cultural adaptation, the gap between DeepL and Claude was 0.76 points, which was the biggest among the five dimensions. In terminology accuracy, GPT performed best, while DeepL was weaker. This is also in line with the case analysis later.

Claude ranked first in both accuracy and style matching, which suggests that it was relatively stable in understanding source meaning and keeping source style. Gemini’s overall average score was slightly lower than GPT and Claude, but it got the highest score in fluency, which shows strong target language generation ability.

**4.4 Heatmap of Translation Quality Distribution**

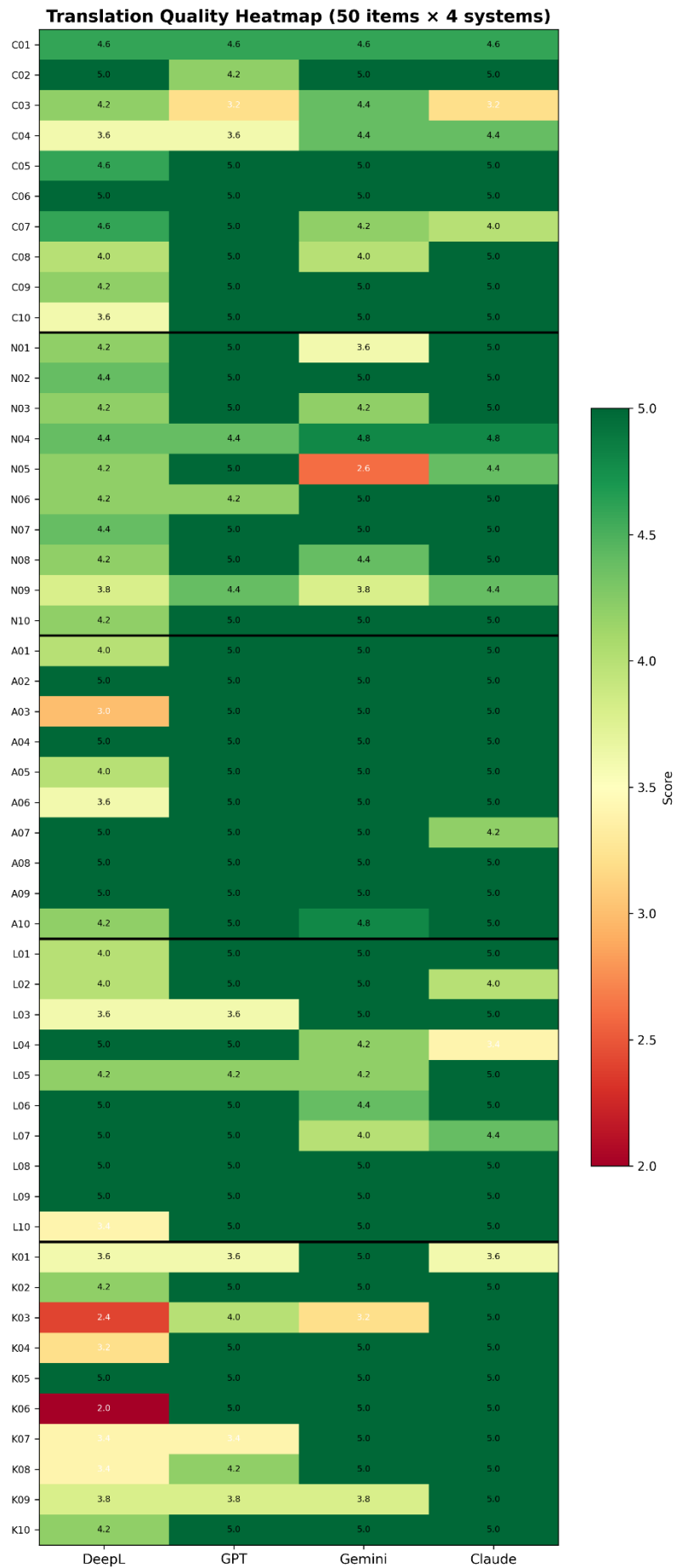


Figure 4. Heatmap of

translation quality scores (50 items × 4 systems).

The heatmap shows the quality distribution of the 200 translations in a direct way. High-score areas were mainly found in academic texts and some structurally simple sentences. This shows that in more standardized and regular texts, all four systems performed quite steadily. Low-score areas were mainly found in the DeepL column for culture-loaded texts, forming a clear low-score area, especially in cases involving two-part allegorical sayings, online abbreviations, and internet slang.

At the same time, the heatmap also shows that even the LLM systems that performed well overall still had low-score points in some special colloquial expressions. For example, the sentence C03, “我要我觉得，不要你觉得,” got low scores in both Claude and GPT. This suggests that for expressions that depend heavily on context and carry strong power relations and pragmatic meaning, all systems still face some difficulties.

### 4.5 Typical Case Analysis

#### Case 1: Translation of YYDS (K06) – DeepL’s Factual Error

**Table 7 Comparison of K06 Translations**

| System | Translation                                                          | Score |
|--------|----------------------------------------------------------------------|-------|
| DeepL  | "which stands for 'You're the God'"                                  | 2.0   |
| GPT    | "meaning 'the GOAT' (the greatest of all time)"                      | 5.0   |
| Gemini | "which means 'the GOAT'"                                             | 5.0   |
| Claude | "which stands for '永远的神' — meaning 'the GOAT' or 'a legend forever'" | 5.0   |

In this case, DeepL explained “YYDS” as “You’re the God,” which is a clear factual mistranslation. In fact, “YYDS” comes from the pinyin initials of “永远的神.” The three LLM systems all recognized the real meaning of this internet term more accurately and gave a close English expression, “GOAT.” Among them, Claude not only kept the original Chinese information, but also added an explanation, which is more helpful for target readers.

#### Case 2: The Term “Irrelevant Speech Effect” (A03) – DeepL’s Terminology Mistranslation

**Table 8 Comparison of Key Term in A03**



| System | Key Term                     | Score |
|--------|------------------------------|-------|
| DeepL  | "non-verbal effects" ❌       | 3.0   |
| GPT    | "irrelevant speech effect" ✅ | 5.0   |
| Gemini | "irrelevant speech effect" ✅ | 5.0   |
| Claude | "irrelevant speech effect" ✅ | 5.0   |

“无关言语效应” is a fixed term in psycholinguistics, and its English form is “irrelevant speech effect.” DeepL translated it as “non-verbal effects,” which is not only wrong as a term, but also shifts the meaning. The three LLM systems all gave the correct term. This suggests that they have a clear advantage in identifying and using professional terminology.

### Case 3: Translation of a Two-part Allegorical Saying (K03) – Claude’s Multi-layer Handling

**Table 9 Comparison of K03 Translations**

| System | Translation                                                                                          | Score |
|--------|------------------------------------------------------------------------------------------------------|-------|
| DeepL  | "A nephew holding a lantern—it still shines on his uncle."                                           | 2.4   |
| GPT    | "A nephew carrying a lantern—just like before (a pun on 'uncle')."                                   | 4.0   |
| Gemini | "Like a nephew carrying a lantern—everything remains the same as always."                            | 3.2   |
| Claude | "A nephew carrying a lantern — same as before (lighting the way for his uncle / business as usual)." | 5.0   |

The difficulty of “外甥打灯笼——照旧（舅）” lies in the homophonic pun between “照旧” and “照舅.” DeepL stayed mostly at the literal level and did not convey the real meaning. Gemini conveyed the meaning of “same as before,” but did not show the pun mechanism. GPT noticed the pun, but the explanation was still not complete enough. Claude kept the literal image, the sound relation, and the actual meaning at the same time, so it got the highest score.

### Case 4: Lu Xun’s “吃人” (L02) – Keeping the Literary Shock Effect

Table 10 Comparison of Key Expression in L02

| System | Key Expression | Score |
|--------|----------------|-------|
| DeepL  | "cannibalism"  | 4.0   |
| GPT    | "eat people"   | 5.0   |
| Gemini | "Eat people"   | 5.0   |
| Claude | "cannibalism"  | 4.0   |

In Lu Xun’s *Diary of a Madman*, the power of the phrase “吃人” lies in its directness and shock effect. GPT and Gemini used “eat people,” which is closer to the style of the source text and keeps more of the literary pressure. By contrast, DeepL and Claude used “cannibalism.” The meaning is not wrong, but the style becomes more formal and abstract, which weakens the emotional effect of the original.

Case 5: Rhetorical Parallelism (K09) – Claude’s Reconstructed Expression

Table 11 Comparison of K09 Translations

| System | Translation                                         | Score |
|--------|-----------------------------------------------------|-------|
| DeepL  | "Not very hurtful, but extremely insulting."        | 3.8   |
| GPT    | "Not very harmful, but extremely insulting."        | 3.8   |
| Gemini | "It's not that harmful, but it's highly insulting." | 3.8   |
| Claude | "The damage is minimal; the insult is devastating." | 5.0   |

The phrase “伤害性不高，侮辱性极强” has a strong rhythm and parallel structure. The first three systems basically conveyed the meaning, but the sentence pattern was rather flat and did not keep the rhetorical effect very well. Claude reorganized the expression and built a stronger contrast and tone in English, so it performed better in this case.

4.6 Error Type Statistics

Table 12 Statistics of Error Types

| Error Type                    | Frequency | Main Systems Involved | Typical Case                                    |
|-------------------------------|-----------|-----------------------|-------------------------------------------------|
| Terminology errors            | 6         | mainly DeepL          | A03 “无关言语效应” → “non-verbal effects”             |
| Cultural loss                 | 12        | DeepL > GPT > Gemini  | K03 pun in the allegorical saying lost          |
| Register shift                | 5         | all systems           | L02 “吃人” → “cannibalism”                        |
| Semantic distortion           | 3         | GPT/Claude            | over-free translation in some cases             |
| Metaphor loss                 | 3         | mainly DeepL          | N06 metaphor of “浪花” lost                       |
| Information addition/deletion | 2         | scattered             | adding or leaving out information in some cases |

Looking at the error distribution, cultural loss was the most common problem, and it was mainly found in DeepL. Terminology errors also appeared mostly in DeepL outputs. Compared with this, the LLM systems performed better overall, but they were not completely without problems. For example, in literary expression and strongly pragmatic sentences, register shift and over-free translation still appeared.

## Chapter Five Discussion

### 5.1 Advantages of LLMs: Knowledge Storage and Pragmatic

#### Understanding

In this study, the LLM systems showed clear advantages in culture-loaded texts and expressions with complex pragmatics. One possible reason is that large language models are exposed to many different types of texts during training, so they are

stronger in cultural knowledge, internet expressions, and contextual inference. Taking “YYDS” as an example, the three LLMs not only recognized its real meaning, but also found a close cultural equivalent in English. Traditional NMT systems find this much harder to do. This is also quite consistent with Guerreiro et al. (2023), who discussed hallucination-like problems in NMT systems.

In colloquial and ironic expressions, LLMs also showed stronger pragmatic understanding. For example, an expression like “我真是谢谢你了” looks polite on the surface, but it actually carries irony. LLMs are more likely to catch the real tone instead of only translating the literal meaning. This suggests that besides word-level and sentence-level transfer, understanding pragmatics also has a strong effect on translation quality.

## **5.2 Relationship Between System Choice and Text Type**

The results show that the strengths of different systems are not exactly the same. GPT performed best in academic texts, which suggests that it is more stable in terminology handling and logical expression. Gemini got the highest score in colloquial texts, which reflects some strength in natural spoken language generation. Claude did especially well in culture-loaded texts and news texts, and it was relatively stable in cultural explanation and style control.

This also means that in real use, translation tools should be chosen with the text type in mind. If the task is academic abstracts or paper passages with stronger regularity, GPT may be a better first choice. If the text contains more culture-loaded content, Claude may be more suitable. If the text is mainly daily expression or spoken content, Gemini is also quite competitive. Although DeepL’s overall score was lower, it still has practical value in the fast translation of standardized and large-volume texts.

## **5.3 The “Simple Text Convergence” Phenomenon**

From the results, academic texts generally got high scores. This means that in sentences with clearer structure and more fixed terminology, the systems are already quite close to each other. In other words, for texts that are not very difficult and that follow regular expression patterns, current machine translation systems can usually give fairly good results, and the differences between systems are not very obvious.

On the other hand, the texts that really separate system quality are often those with cultural embedding, rhetorical effects, strong pragmatic color, or high demands for professional terminology. This suggests that in future machine translation evaluation, looking only at the overall average score may not be enough. It is also necessary to combine this with the analysis of specific difficulties and error types if we want to judge system ability more accurately.

## **5.4 Fluency Differences Are Smaller, but Cultural Handling Is Still Difficult**

In this study, the fluency dimension had the smallest score gap. This suggests that “writing smooth English” is no longer the hardest problem for machine translation systems. What is really harder now is how to keep cultural meaning, register style, and expressive effect while still staying fluent.

This also gives some useful ideas for translation practice. For post-editing, the focus in the future may not only be grammar correction and sentence polishing. More attention may need to go to cultural meaning, rhetorical effect, and pragmatic appropriateness. For evaluation research, if fluency is still treated as the most central index, it may no longer be enough to separate system ability effectively.

## **5.5 LLMs’ Ability to Reconstruct Expression**

From the case analysis, we can see that some LLM systems are not only doing word-for-word translation anymore. In some situations, they actively reconstruct the expression in order to keep the source tone, rhetoric, or style as much as possible. Claude’s handling of K09 is a typical example. It did not strictly follow the source sentence pattern. Instead, it reorganized the structure in English to get closer to the effect of the source text.

This suggests that in some cases, LLMs already show a certain degree of effect-oriented translation ability. This ability is no longer fully limited to what Nida (1964) described as formal equivalence or dynamic equivalence. It also has some features of creative reconstruction. These systems are not only paying attention to what the source text says, but also, to some degree, to how it says it and what kind of feeling it gives. Of course, this kind of reconstruction is not always successful. Sometimes it can also lead to over-free translation, so human judgment and checking are still necessary.

## **5.6 Strengths and Limits of DeepL**

Although DeepL ranked last in this study overall, it still has its own strengths. First, DeepL is easy to use and fast, so it is suitable for quickly getting a basic translation. Second, in texts with clearer structure and average difficulty, DeepL’s output is still usable to some degree.

However, its limits are also obvious. The first is that terminology handling is not stable enough. When it meets professional terms that are not well covered in training, mistranslation is more likely to appear. The second is that its cultural adaptation ability

is weaker. For two-part allegorical sayings, internet slang, and cultural allusions, it often cannot give very ideal results. So if a text contains strong cultural meaning or professional knowledge, DeepL's output needs more human checking.

## **5.7 Limitations of the Study**

This paper also has some limits.

First, the scoring in this study was completed by only one evaluator, so some subjectivity is unavoidable. Although the scoring process included note-taking and checking again, the results would be more convincing if future studies include several evaluators and report inter-rater consistency.

Second, the data size is still small. Although the 50 texts cover five text types, each type only includes 10 items, so they cannot fully represent all the internal complexity of that type. Academic texts especially only involve some fields at present. In the future, this can be extended to medicine, law, economics, and other areas.

Third, this study only looks at one direction, Chinese to English. The structural and cultural differences between Chinese and English are large, and this may enlarge some system differences. Future studies that include other language pairs may give a more complete picture.

Fourth, system versions keep changing. The data in this study were collected on April 18, 2025, so the results should be understood more as a horizontal comparison at that time, not as a permanent conclusion about system ability.

Fifth, earlier studies have pointed out that prompt design can have a visible influence on the translation quality of LLMs (Peng et al., 2023). This paper uses a zero-prompt setting and does not give special prompt optimization to the LLMs. This helps control variables, but it may also underestimate their potential in real translation situations.

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# **Chapter Six Conclusion**

## **6.1 Main Findings**

Based on 50 Chinese source texts and 200 translations, this paper compared the Chinese-to-English translation quality of four machine translation systems: DeepL, ChatGPT (GPT-4o), Google Gemini, and Claude. Through five-dimensional human evaluation, the following conclusions can be drawn.

First, the three LLM systems all performed better than DeepL overall. Claude got the highest average score, followed by GPT and Gemini. This suggests that within the range of texts included in this study, large language model systems already show stronger overall ability in Chinese-to-English translation than traditional NMT.

Second, text type has a strong influence on translation quality. In academic texts, the systems performed similarly. In culture-loaded texts, the differences were much clearer. This means that the real quality gap between systems mainly appears in more difficult text types.

Third, fluency is no longer the most central dimension for separating system ability. Cultural adaptation and terminology accuracy show the system differences more clearly. In other words, current machine translation systems can usually generate fairly natural English, but clear differences still remain in cultural transfer and professional expression accuracy.

Fourth, each system has its own strong area. GPT is stronger in academic texts, Gemini sounds more natural in colloquial expression, and Claude performs better in culture-loaded texts and style control. So in real use, the system should be chosen flexibly according to the text type, instead of depending on only one tool.

Fifth, some advanced LLMs have already shown a certain ability to reconstruct expression, and they can try to rebuild the rhetorical effect and expressive force of the source text in the target language. Still, this ability is not always stable, so human post-editing remains necessary.

## **6.2 Practical Suggestions**

Based on the findings above, this paper gives several suggestions.

First, if the text is mainly academic, GPT can be considered first, because it is relatively stable in terminology handling and accurate expression.

Second, if the text contains much culture-loaded information, such as internet slang, two-part allegorical sayings, or allusions, Claude is often a better choice.

Third, if the text is more colloquial, Gemini has some advantage in natural expression.

Fourth, if the task is only to get a quick first draft of a standardized text, DeepL can still be used as an efficient tool, but later human checking is still suggested.

Fifth, no matter which system is chosen, human revision is still necessary, especially in texts with high literary, cultural, or professional demands.

## **6.3 Suggestions for Future Research**

Future studies can continue in several directions.

First, the data size and text range can be expanded by including more fields and more real application settings, so the results can be more representative.

Second, more evaluators can be invited to score the data, and inter-rater consistency can be reported to improve the reliability of the study.

Third, longitudinal research can be carried out to track how different systems change after version updates.

Fourth, future studies can examine how prompt design affects the translation quality of LLMs, and compare strategies such as zero-shot prompting, few-shot prompting, and role prompting.

Fifth, human evaluation and automatic evaluation can be combined to explore a more efficient mixed evaluation method for Chinese-to-English translation tasks.

Sixth, researchers can further study the possibility of multi-system collaborative translation and see whether combining different systems can improve overall translation quality.

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# Appendix

**Appendix A:** Full list of the 50 Chinese source texts (see Table 1)

**Appendix B:** Full table of 200 translation outputs and scoring data (see Excel file *LLM\_Translation\_Evaluation.xlsx*)

**Appendix C:** Four data visualization figures (*fig1\_overall.png*, *fig2\_by\_type.png*, *fig3\_radar.png*, *fig4\_heatmap.png*)